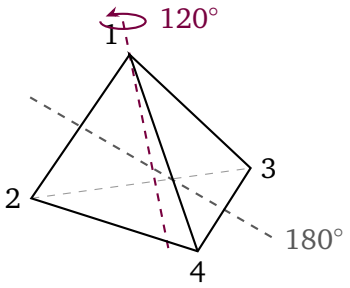
# Symmetries in Physics — Exercise Sheet 5

*Characters and character tables*

**Variant: Questions only**

**Exercise 1.** Construct the character table of $A_4$, the rotation group of the tetrahedron. The picture, taken from Lecture 2, shows one rotation axis of each kind:




Here is everything you will need (Lecture 2's example $A_4/V \cong C_3$).

- $|A_4| = 12$, and $A_4$ has four conjugacy classes: the identity; the three half-turns $(1\,2)(3\,4)$, $(1\,3)(2\,4)$, $(1\,4)(2\,3)$; the four 3-cycles $(1\,2\,3)$, $(1\,3\,4)$, $(1\,4\,2)$, $(2\,4\,3)$; and the four inverse 3-cycles $(1\,3\,2)$, $(1\,4\,3)$, $(1\,2\,4)$, $(2\,3\,4)$.
- The identity and the three half-turns form a normal subgroup $V$, and $A_4/V \cong C_3$. The projection $\pi\colon A_4 \to C_3$ sends $V$ to $e$, the class of 3-cycles containing $(1\,2\,3)$ to $r$, and the class of $(1\,3\,2)$ to $r^2$.
- The characters of $C_3$ are $\chi_k(r^m) = \omega^{km}$, with $\omega = e^{2\pi \mathrm{i}/3}$ and $k = 0, 1, 2$.

(i) Without constructing any representation, use the two counting theorems of Lecture 5 to find the dimensions of the irreducible representations of $A_4$. Do the same for $D_6$, which has order 12 and six conjugacy classes (Sheet 2, Exercise 2).

(ii) Construct three one-dimensional representations of $A_4$ as $\chi_k \circ \pi$.

(iii) Find the missing row of the table from $\sum_i n_i \chi_i(g) = 0$ for $g \neq e$, and check its norm.

**Exercise 2.** Lecture 5's example on $D_4$ claimed that the quaternion group $Q_8$ has the same character table. Prove it. Here is everything you will need (Sheet 4, Exercise 6).

- $Q_8 = \{\pm 1, \pm \mathrm{i}, \pm \mathrm{j}, \pm \mathrm{k}\}$, with $\mathrm{i}^2 = \mathrm{j}^2 = \mathrm{k}^2 = \mathrm{ijk} = -1$. Any two of $\mathrm{i}, \mathrm{j}, \mathrm{k}$ anticommute, for instance $\mathrm{ij} = \mathrm{k} = -\mathrm{ji}$.
- Four one-dimensional representations $\psi_{ab}$, with $a, b = \pm 1$:

$$\psi_{ab}(\pm 1) = 1, \quad \psi_{ab}(\pm \mathrm{i}) = a, \quad \psi_{ab}(\pm \mathrm{j}) = b, \quad \psi_{ab}(\pm \mathrm{k}) = ab.$$

- A two-dimensional irreducible representation

$$\mathrm{i} \mapsto -\mathrm{i}\sigma_1, \quad \mathrm{j} \mapsto -\mathrm{i}\sigma_2, \quad \mathrm{k} \mapsto -\mathrm{i}\sigma_3, \quad -1 \mapsto -\mathbb{1}.$$

Here $\sigma_1 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\sigma_2 = \left(\begin{smallmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{smallmatrix}\right)$, $\sigma_3 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ are the Pauli matrices.

- The table of $D_4$ (Lecture 5):

| $D_4$ | $1\,e$ | $1\,r^2$ | $2\,r$ | $2\,s$ | $2\,rs$ |
|---|---|---|---|---|---|
| $A_1$ | 1 | 1 | 1 | 1 | 1 |
| $A_2$ | 1 | 1 | 1 | −1 | −1 |
| $B_1$ | 1 | 1 | −1 | 1 | −1 |
| $B_2$ | 1 | 1 | −1 | −1 | 1 |
| $E$ | 2 | −2 | 0 | 0 | 0 |

  The lecture labels these rows $\chi_1, \dots, \chi_5$; here we use Mulliken's symbols.

(i) Find the conjugacy classes of $Q_8$ and their sizes.

(ii) Prove that these five representations are all the irreducible representations of $Q_8$.

(iii) Write down the character table of $Q_8$ and compare it with that of $D_4$.

(iv) Prove that $D_4 \not\cong Q_8$. What does this say about the character table?

**Exercise 3.** $S_4$ permutes the coordinates of $\mathbb{C}^4$. Compute the character of this permutation representation, then use the character calculus to decompose it and to prove that its three-dimensional summand is irreducible.

**Exercise 4.** Let $\rho$ be a $d$-dimensional representation of a finite group $G$, and $\chi$ its character.

(i) Show that $|\chi(g)| \le d$ for every $g$, with equality if and only if $\rho(g)$ is a scalar multiple of the identity. Deduce that $\chi(g) = d$ exactly on the kernel of $\rho$.

(ii) Suppose that $\rho$ is faithful. Prove that $|\chi(g)| = d$ only for central elements $g$.

(iii) From the tables of Exercise 2, find which irreducible representations of $D_4$ and $Q_8$ are faithful, and at which elements $|\chi(g)| = d$.